

RONGYAO FANG

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EDUCATION

The Chinese University of Hong Kong

Sept. 2021 - Oct. 2025

Ph.D. at MMLab, Department of Electronic Engineering.

Supervisor: Prof. Hongsheng Li. Also worked closely with Prof. Xihui Liu.

Topics: Multimodal Large Language Models, Artificial General Intelligence.

Shanghai Jiao Tong University

Sept. 2016 - July 2020

B.Eng., School of Electronic Information and Electrical Engineering.

Major: Information Engineering (Artificial Intelligence track).

Ranking: 1st/157.

Research: Independent researcher under the supervision of Prof. Bingbing Ni.

Massachusetts Institute of Technology

July 2019 - March 2020

Computer Science and Artificial Intelligence Laboratory (CSAIL).

Research: Independent visiting scholar under the supervision of Prof. Dina Katabi .

AREAS OF EXPERTISE

My work focuses on **vision-language agents for real-world productivity** – deploying long-horizon multimodal agents to automate high-value design and engineering workflows such as CAD modeling, 3D asset production in Blender, and UI/UX design. I also lead efforts on enhancing LLMs' **web development capabilities**, including AI-driven front-end code generation and vision-to-code. Driven by a vision of **Artificial General Intelligence (AGI)**, my goal is to bring vision-language models into real-world production tasks to tangibly boost human productivity.

EXPERIENCE

Alibaba Qwen VL Team

Sept. 2025 - Present

Role: Research Scientist. **Topics:** AI-powered web/app development, multimodal coding agents.

SenseTime

Feb. 2024 - Aug. 2025

Role: Research Intern. **Topics:** Unified multimodal large language models.

Shanghai AI Laboratory

June 2022 - Apr. 2023

Role: Research Intern. **Topics:** Representation learning and vision perception.

SELECTED PROJECTS

Vision-Language Agents for Real-World Productivity

- Leading the development of **long-horizon multimodal agents** for real-world productivity scenarios at Alibaba Qwen VL Team. Building agentic systems that autonomously operate professional tools (CAD, Blender, design software) to automate **3D asset production, architectural modeling, and UI/UX design** workflows.

AI-Powered Web Development

- Core researcher on enhancing Qwen models' web development capabilities: enabling LLMs to generate **production-ready front-end code and interactive web applications** from natural language, and building multimodal coding agents for vision-to-code generation.

Reasoning-Enhanced Visual Generation (GoT Series)

- Led the GoT project series on integrating **chain-of-thought reasoning into multimodal LLMs for visual generation and editing**. GoT-R1 further introduced **reinforcement learning** to unlock reasoning capabilities. Both accepted to top venues (NeurIPS 2025, ICLR 2026).

The Verification Horizon: No Silver Bullet for Coding Agent Rewards

Binghai Wang, Chenlong Zhang, Dayiheng Liu, Jiajun Zhang, Jiawei Chen, Mouxiang Chen, **Rongyao Fang**, Siyuan Zhang, Xuwu Wang, Yuheng Jing, Zeyao Ma, Zeyu Cui. (alphabetical order) Qwen Team Official Technical Report, 2026 ([Link](#)).

SpecV: Specification Verification for Robust Unified Multimodal Evaluation

Weihao Yu*, **Rongyao Fang***, Yuxuan Cai*, et al. European Conference on Computer Vision (**ECCV 2026**).

MathCanvas: Intrinsic Visual Chain-of-Thought for Multimodal Mathematical Reasoning

Weihan Shi*, Aldrich Yu*, **Rongyao Fang***, et al. Association for Computational Linguistics (**ACL 2026**) ([Link](#)).

CodePlot-CoT: Mathematical Visual Reasoning by Thinking with Code-Driven Images

Chengqi Duan*, Kaiyue Sun*, **Rongyao Fang***, et al. Computer Vision and Pattern Recognition (**CVPR Findings 2026**) ([Link](#)). **Best Paper Award**, KnowledgeMR Workshop, CVPR 2026.

FLUX-Reason-6M & PRISM-Bench: A Million-Scale Text-to-Image Reasoning Dataset and Benchmark

Rongyao Fang, Aldrich Yu, Chengqi Duan, et al. International Conference on Learning Representations (**ICLR 2026**) ([Link](#)).

GoT-R1: Unleashing Reasoning Capability of MLLM for Visual Generation with Reinforcement Learning

Chengqi Duan*, **Rongyao Fang***, Yuqing Wang*, et al. International Conference on Learning Representations (**ICLR 2026**) ([Link](#)).

Qwen3-VL Technical Report

Official Technical Report of Qwen3-VL, Alibaba Group, 2025 ([Link](#)).

GoT: Unleashing Reasoning Capability of MLLM for Visual Generation and Editing

Rongyao Fang, Chengqi Duan, Kun Wang, et al. Conference on Neural Information Processing Systems (**NeurIPS 2025**) ([Link](#)).

PUMA: Empowering Unified MLLM with Multi-Granular Visual Generation

Rongyao Fang, Chengqi Duan, Kun Wang, et al. International Conference on Computer Vision (**ICCV 2025**) ([Link](#)).

FouriScale: A Frequency Perspective on Training-Free High-Resolution Image Synthesis

Linjiang Huang*, **Rongyao Fang***, et al. European Conference on Computer Vision (**ECCV 2024**) ([Link](#)).

FeatAug-DETR: Enriching One-to-Many Matching for DETRs with Feature Augmentation

Rongyao Fang, Peng Gao, Aojun Zhou, et al. Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**) ([Link](#)).

Tip-Adapter: Training-Free CLIP-Adapter for Better Vision-Language Modeling

Renrui Zhang*, **Rongyao Fang***, Peng Gao*, et al. European Conference on Computer Vision (**ECCV 2022**) ([Link](#)).

CLIP-Adapter: Better Vision-Language Models with Feature Adapters

Peng Gao, Shijie Geng, Renrui Zhang, Teli Ma, **Rongyao Fang**, et al. International Journal of Computer Vision (**IJCV**) ([Link](#)).

Learning Longterm Representations for Person Re-Identification Using Radio Signals

Lijie Fan*, Tianhong Li*, **Rongyao Fang***, et al. IEEE Conference on Computer Vision and Pattern Recognition (**CVPR 2020**) ([Link](#)).

HONORS AND AWARDS

Hong Kong PhD Fellowship

Research Grants Council (RGC) of Hong Kong.

Sept. 2021

Outstanding Graduates of Shanghai*July 2020*

TOP 1%, Shanghai Municipal Education Commission.

National Scholarship*2017 & 2018*

TOP 1%, Ministry of Education of P.R.China.

Zhiyuan College Honors Scholarship*2017 & 2018*

TOP 5%, Zhiyuan College, Shanghai Jiao Tong University.